

SchoolDesk user flow diagram and QA test case list covering Admin, Principal, Teacher, and Parent RBAC roles

User Flow	Test Cases (80)				
Search by ID, title, or keyword...			All roles	All modules	All types
Showing 80 of 80 test cases					
ID	Role	Module	Test Case	Expected Result	Type
TC001	All	Auth	Valid login — Admin role	Admin dashboard loads with KPI overview (students, teachers, attendance)	Positive
TC002	All	Auth	Valid login — Principal role	Principal dashboard loads with school-wide attendance & performance overview	Positive
TC003	All	Auth	Valid login — Teacher role	Teacher dashboard loads showing own class name, student count, today's attendance status	Positive
TC004	All	Auth	Valid login — Parent role	Parent dashboard loads with linked child's name, class, and today's attendance	Positive
TC005	All	Auth	Login with incorrect password	Toast error: "Invalid credentials" — user stays on login screen, no account details revealed	Negative
TC006	All	Auth	Login with empty email field	Inline validation: "Email is required" before API call is made	Negative
TC007	All	Auth	Login with empty password field	Inline validation: "Password is required" before API call is made	Negative
TC008	All	Auth	Login with non-existent email	"Invalid credentials" shown — same message as wrong password (no account enumeration)	Negative
TC009	All	Auth	Session JWT expires after idle timeout	User redirected to login with message "Session expired. Please log in again."	Edge Case
TC010	All	Auth	Logout clears session and invalidates JWT	Token blacklisted server-side; navigating back to any route redirects to login	Positive
TC011	Admin	User Management	Admin dashboard loads with all KPI cards	Shows total students, teachers, parents, today's school attendance %, pending fee	Positive

ID	Role	Module	Test Case	Expected Result	Type
				count	
TC012	Admin	User Management	Add new teacher with all required fields	Teacher account created; confirmation toast shown; teacher can log in immediately	Positive
TC013	Admin	User Management	Add teacher with an already-registered email	Error: "A user with this email already exists"	Negative
TC014	Admin	User Management	Edit existing teacher's profile (name, phone, subject)	Updated data saved; changes reflected across all views immediately	Positive
TC015	Admin	User Management	Deactivate (soft-delete) a teacher account	Teacher marked inactive; teacher login returns "Account disabled. Contact admin."	Positive
TC016	Admin	User Management	Add new parent and link to an existing student	Parent created and linked; parent dashboard shows correct child's data	Positive
TC017	Admin	User Management	Add new student and assign to a class	Student created and appears in class student list; visible in teacher's class view	Positive
TC018	Admin	User Management	Add student without required fields (name, class)	Form validation errors for all missing required fields; no API call made	Negative
TC019	Admin	User Management	Bulk import students via CSV file	Valid rows imported successfully; invalid rows highlighted in an error report download	Edge Case
TC020	Admin	User Management	Search/filter user list by name or class	Results filter in real-time matching search query across name and class fields	Positive
TC021	Admin	School Setup	Create new class with section (e.g. Class 5 — A)	Class/section created; appears in class dropdowns and teacher assignment	Positive
TC022	Admin	School Setup	Create a new academic year and set as active	New year activated; previous year's data archived and still accessible as history	Positive
TC023	Admin	School Setup	Assign teacher to a class	Teacher now sees that class in their dashboard; previous class assignment removed if exclusive	Positive
TC024	Admin	School Setup	Define fee structure for a class	Fee amounts saved; parents of that class see correct fee breakdown on dashboard	Positive
TC025	Admin	School Setup	Update school name and logo in School Settings	Changes reflected in app header and all generated reports immediately	Positive

ID	Role	Module	Test Case	Expected Result	Type
TC026	Admin	Notices	Upload class routine PDF for a specific class	Routine visible to assigned teacher and parents of that class only	Positive
TC027	Admin	Notices	Upload exam schedule for a class	Schedule visible to all roles under exam schedule section with correct dates	Positive
TC028	Admin	Notices	Create and publish school-wide notice	Notice appears on notice board for all roles immediately after publish	Positive
TC029	Admin	Notices	Create notice with missing title field	Validation error: "Title is required" — notice not saved	Negative
TC030	Admin	RBAC & Security	Admin tenant isolation — access another school's data	All cross-tenant API requests return 403 Forbidden with no data leakage	Edge Case
TC031	Admin	Reports	Generate school-wide attendance report	Report shows per-class attendance % with date range filter working correctly	Positive
TC032	Admin	Reports	Generate academic performance report	Shows class-wise and subject-wise marks, pass/fail rates correctly	Positive
TC033	Admin	Reports	Generate fee collection report	Shows paid, pending, overdue totals with student-level breakdown	Positive
TC034	Admin	Reports	Export report as PDF	PDF downloads with correct header, data, and formatting. No placeholder data.	Positive
TC035	Admin	Reports	Generate report for date range with no data	Empty state message: "No records found for the selected date range"	Edge Case
TC036	Principal	Reports	Principal dashboard shows school-wide KPIs	Total enrollment, avg attendance %, academic avg, and fee status visible	Positive
TC037	Principal	Attendance	View school-wide attendance summary	Per-class daily attendance breakdown shown; low-attendance classes highlighted	Positive
TC038	Principal	Reports	View class-wise academic performance report	Class ranks and subject averages displayed accurately	Positive
TC039	Principal	Reports	View teacher activity log	Log shows attendance marking timestamps, homework creation, marks entry per teacher	Positive
TC040	Principal	Notices	Approve a pending notice submitted by teacher/admin	Notice status changes to "Approved" and becomes visible on notice board	Positive

ID	Role	Module	Test Case	Expected Result	Type
TC041	Principal	Notices	Reject a notice with a reason	Notice status set to "Rejected"; creator receives rejection notification with reason	Positive
TC042	Principal	Reports	View fee collection report	Correct totals, class-wise breakdown, and pending amounts displayed	Positive
TC043	Principal	Reports	Export attendance report as PDF/Excel	Exported file contains accurate, complete data matching on-screen view	Positive
TC044	Principal	RBAC & Security	Principal tries to create or edit user accounts	User management section is not accessible; attempting API call returns 403	Negative
TC045	Principal	RBAC & Security	Principal scoped to own school tenant only	Cannot view or report on students, teachers, or data from other schools	Edge Case
TC046	Teacher	Attendance	Teacher dashboard shows own class info on load	Class name, total students, today's attendance status (pending/done) visible	Positive
TC047	Teacher	Attendance	Mark all students present for today	Attendance saved; visible in admin and principal attendance reports	Positive
TC048	Teacher	Attendance	Mark individual student as absent	Student marked absent; parent dashboard shows absent status for today	Positive
TC049	Teacher	Attendance	Edit today's attendance after initial submission	Correction allowed on same calendar day; change logged with edit timestamp	Edge Case
TC050	Teacher	Attendance	Attempt to mark attendance for a future date	Future date selector disabled or blocked; error: "Cannot mark future attendance"	Negative
TC051	Teacher	Attendance	View monthly attendance report for own class	Calendar heat-map view shows daily present/absent counts correctly for the month	Positive
TC052	Teacher	Homework	Create new homework assignment with subject and deadline	Assignment appears in parent and student homework lists with correct deadline	Positive
TC053	Teacher	Homework	Create assignment with a deadline in the past	Warning or block shown: "Deadline cannot be in the past"	Edge Case
TC054	Teacher	Homework	View all submissions for a homework assignment	List shows submitted / pending / late status per student accurately	Positive
TC055	Teacher	Homework	Grade a student's homework submission	Grade saved; visible to parent and reflected in student's academic summary	Positive

ID	Role	Module	Test Case	Expected Result	Type
TC056	Teacher	Homework	Create homework assignment with file attachment	File uploaded and accessible to parents/students via download link	Positive
TC057	Teacher	RBAC & Security	Teacher tries to view students from another class	Only own class students visible in all lists; other students' data inaccessible	Edge Case
TC058	Teacher	Exams & Results	Create exam schedule for own class	Exam appears on schedules visible to admin, principal, parents, and students	Positive
TC059	Teacher	Exams & Results	Enter marks for all students in a subject	Marks saved; results appear in reports after teacher publishes	Positive
TC060	Teacher	Exams & Results	Enter marks exceeding the configured maximum marks	Validation error: "Marks cannot exceed [max marks]" — entry blocked	Negative
TC061	Teacher	Notices	Post a notice to own class only	Notice visible only to students/parents of that class; not school-wide board	Positive
TC062	Parent	Attendance	Parent dashboard loads with correct child summary	Child's name, class, today's attendance status, and latest homework shown	Positive
TC063	Parent	Attendance	View child's monthly attendance record	Calendar shows present/absent/holiday per day correctly for the month	Positive
TC064	Parent	Homework	View child's active homework list	All assignments with subjects, deadlines, and submission status displayed	Positive
TC065	Parent	Homework	Submit homework on behalf of child	Submission recorded; teacher sees it in submission list with parent-submitted tag	Positive
TC066	Parent	Homework	Submit homework after the deadline	Submission accepted but flagged as late; teacher sees late submission indicator	Edge Case
TC067	Parent	Exams & Results	View child's upcoming exam schedule	Correct dates, subjects, and timings shown in chronological order	Positive
TC068	Parent	Exams & Results	View child's published exam results	Subject-wise marks and total shown only after teacher has published results	Positive
TC069	Parent	Exams & Results	View results before teacher publishes them	Results section shows "Results not yet published" — no marks visible	Edge Case
TC070	Parent	Notices	View school-wide and class-specific notices	Both school notices and class-level notices visible; correctly tagged by scope	Positive

ID	Role	Module	Test Case	Expected Result	Type
TC071	Parent	School Setup	View class routine for child's class	Weekly timetable with subjects and timings displayed correctly	Positive
TC072	Parent	School Setup	View fee payment status for child	Outstanding fees, paid receipts, and due dates shown accurately	Positive
TC073	Parent	RBAC & Security	Parent tries to view another student's profile or data	Only child linked to the parent account is accessible; other students return 403	Edge Case
TC074	All	RBAC & Security	Parent hits an admin-only API route	403 Forbidden returned with message "Insufficient permissions"	Negative
TC075	All	RBAC & Security	Teacher hits a principal-only reports API route	403 Forbidden; teacher cannot access or read any principal report data	Negative
TC076	All	RBAC & Security	API request with tampered role claim in JWT	Backend validates role from DB, not from token claim; 401/403 returned	Edge Case
TC077	All	RBAC & Security	API request to protected route with no auth token	401 Unauthorized on all protected endpoints — no data returned	Negative
TC078	All	RBAC & Security	Password change invalidates all existing sessions	Old JWT tokens rejected after password change; user must log in again	Edge Case
TC079	All	RBAC & Security	Multiple failed logins trigger rate limiting	After 5 failed attempts: account locked for N minutes or CAPTCHA shown	Edge Case
TC080	All	RBAC & Security	SQL injection in login email field	Input sanitized by parameterized query; no SQL error exposed; login fails normally	Edge Case